

Project Functionality

The CS 340 project 2 system is used to compile data from animal shelters and create a data table, pie chart, and geo-map of the data. This system includes the ability to filter and sort the data in the table.

Tools Used

Python: Used to create the scripts in the system

Dash: Used to create the framework of the web application, primarily widgets

MongoDB: Used for storing and retrieving data between the animal shelter and the Python script

Plotly: Used to create charts using the data from the system

Steps to complete the project

First, the MongoDB and the .ipynb code were setup. Then the CRUD system was created as a .py file. Then a dashboard was created in a .ipynb file with login credentials for the aac database. Then the layout of the webpage was created, and call backs were used to edit the webpage as inputs were being read. Then the code was tested and debugged.

Challenges

Some challenges that accorded in the development of this system were the experimentation of the best way to implement the dropdown widget. At first it was just learning to implement it on the webpage. Then it was getting the dropbox to interact with the table to filter using the selected option. Then challenge was overcome with a bit of brute force tactics for trying everything I could think of till it works and then finding out that I had made a syntax error using single quotes to attempt to query using CRUD instead of double quote like I was supposed to use.

Screenshots

```

app = SuppleDash(brand_name)

#FIX ME Add in Grazioso Salvare's Logo
image_filename = 'Grazioso Salvare Logo.png' # replace with your own image
encoded_image = base64.b64encode(open(image_filename, 'rb').read())

#FIX ME Place the HTML image tag in the line below into the app.layout code according to your design
#FIX ME Also remember to include a unique identifier such as your name or date
#html.Img(src='data:image/png;base64,{}'.format(encoded_image.decode()))

app.layout = html.Div([
#    html.Div(id='hidden-div', style={'display':'none'}),
    html.Center(html.B(html.H1('CS-340 Dashboard'))),
    html.Hr(),
    html.Div(

#FIXME Add in code for the interactive filtering options. For example, Radio buttons, drop down, checkboxes, etc.
[
    dcc.Dropdown(df.animal_type.unique(), style={'width':'300px'}, id='animal_type_filter', placeholder='Animal Type'),
    dcc.Dropdown(df.breed.unique(), style={'width':'300px'}, id='breed_filter', placeholder='Breed'),
    dcc.Dropdown(df.color.unique(), style={'width':'300px'}, id='color_filter', placeholder='Color'),
],
    style={'display':'flex', 'flexDirection':'row','labelstyle':'margin-left'}),
    html.Div(),
    html.Hr(),
    dash_table.DataTable(id='datatable-id',
                        columns=[{"name": i, "id": i, "deletable": False, "selectable": True} for i in df.columns],
                        data=df.to_dict('records')),
#FIXME: Set up the features for your interactive data table to make it user-friendly for your client
#If you completed the Module Six Assignment, you can copy in the code you created here
                        editable = False,
                        filter_action = "native",
                        sort_action = "native",
                        column_selectable = "single",
                        row_selectable = "single",
                        ),

#####
@app.callback(Output('datatable-id', 'data'),
              [Input('animal_type_filter', 'value'),
               Input('breed_filter', 'value'),
               Input('color_filter', 'value')])
def update_dashboard(animalTypeFilter, breedFilter, colorFilter):
    ## FIX ME Add code to filter interactive data table with MongoDB queries
    query = {}
    if animalTypeFilter is not None:
        query["animal_type"] = str(animalTypeFilter)
    if breedFilter is not None:
        query["breed"] = str(breedFilter)
    if colorFilter is not None:
        query["color"] = str(colorFilter)

    data = data = pd.DataFrame.from_records(db.read(query))
    data.drop(columns=['_id'], inplace=True)
    return data.to_dict('records')

```

CS-340 Dashboard

Animal Type Breed Color												
rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name			
filter d:												
2	1 year	A725717	Cat	Domestic Shorthair Mix	Silver Tabby	2015-05-02	2016-05-06 10:49:00	2016-05-06T10:49:00				
9	3 years	A720214	Dog	Labrador Retriever Mix	Red/White	2013-02-04	2016-02-11 12:41:00	2016-02-11T12:41:00	Blessing			
10	3 months	A664290	Cat	Domestic Shorthair Mix	Tortie	2013-09-01	2013-12-08 14:58:00	2013-12-08T14:58:00	*Taylor			
5	2 years	A691584	Dog	Labrador Retriever Mix	Tan/White	2012-11-06	2015-05-30 13:48:00	2015-05-30T13:48:00	Luke			

CS-340 Dashboard

Other														Raccoon Mix				Color			
	☐ rec_num	☐ age_upon_outcome	☐ animal_id	☐ animal_type	☐ breed	☐ color	☐ date_of_birth	☐ datetime	☐ monthyear	☐ name	☐ outcome_subtype	☐ outcome_type	☐ sex_upo								
	filter d:																				
☐	70	3 months	A683276	Other	Raccoon Mix	Black/Gray	2014-04-08	2014-07-09 08:52:00	2014-07-09T08:52:00		Rabies Risk	Euthanasia									
☐	178	1 month	A752260	Other	Raccoon Mix	Gray/Black	2017-04-20	2017-06-20 14:06:00	2017-06-20T14:06:00		Medical	Euthanasia									
☐	334	1 year	A675124	Other	Raccoon Mix	Black/Gray	2013-03-23	2014-03-23 17:30:00	2014-03-23T17:30:00		Suffering	Euthanasia									
☐	500	1 year	A675587	Other	Raccoon Mix	Black/Gray	2013-03-29	2014-03-30 14:45:00	2014-03-30T14:45:00		Rabies Risk	Euthanasia									

Preferred Animals

